



Loughborough
University



AI Solutions for Healthcare

Georgina Cosma

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Professor Georgina Cosma

Academic Position

- Professor of AI & Data Science, Loughborough University
- PhD in Computer Science, University of Warwick (2008)

Research Focus

- AI in health and social care
- Natural Language Processing (LLMs)
- Information retrieval (text, multi-modal searching)
- Information processing and modelling
- Continual machine learning and unlearning
- Responsible AI innovation (bias management and mitigation)

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Professor of AI & Data Science

Professor Georgina Cosma

Leading innovative research in Neural Information Retrieval, Computational Intelligence, and Responsible AI at Loughborough University.



About Me

I am Professor of AI & Data Science at Loughborough University, where I lead the Neural Information Processing, Retrieval & Modelling research group. My research focuses on developing AI systems that are not only technically advanced but also ethical, fair, and beneficial to society.

My academic journey began with a PhD in Computer Science from the University of Warwick, specialising in intelligent information retrieval. Since then, I have built a research programme that spans neural information retrieval, computational intelligence, and responsible AI development.

In addition to my research, I teach the Natural Language Processing (NLP) module to MSc Artificial Intelligence and MSc Advanced Computer Science students, focusing on foundational and state-of-the-art topics including vector space models, information retrieval, large language models (LLMs), and their applications.

What drives my research is the conviction that AI must serve humanity responsibly. I work on projects that address real societal challenges—from healthcare inequalities to supporting vulnerable populations.

**Academic Excellence**
Leading research group with distinguished publication record

**Grant Leadership**
Principal and co-investigator on externally funded projects

**Social Impact**
Addressing healthcare inequalities through responsible AI

**Community Service**
Active reviewer for EPSRC and other funding bodies. Associate Editor of Elsevier Pattern Recognition

**Government Partnerships**
Collaborations with NHS trusts and local authorities

**Industry Innovation**
Strategic partnerships with industry leaders

Extracting and Analysing Human Factors and Safety Insights from Maternity Incident Investigation Reports Using Machine Learning

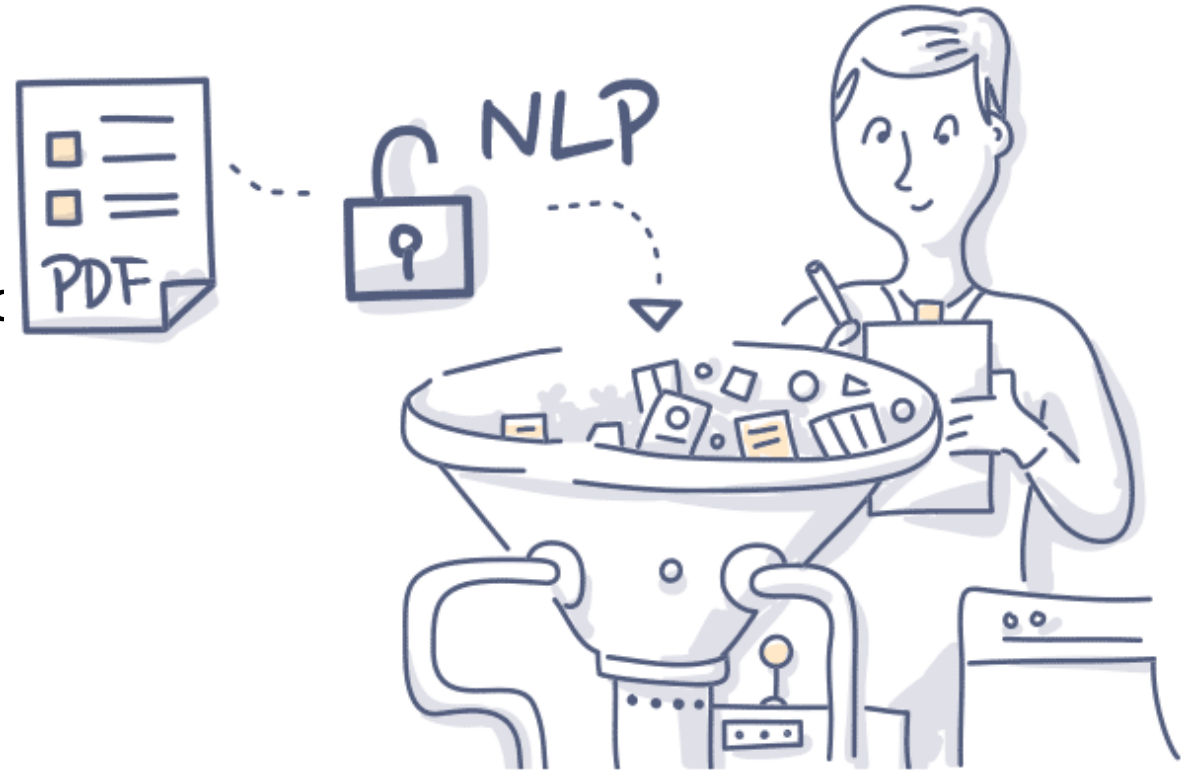
This 2-year project is independent research funded by NHSX and The Health Foundation and it is managed by the National Institute for Health Research (AI_HI200006).
(January 2022-June 2024)

About the Project

HSIB was established in 2017 to improve patient safety through independent investigations

Transformed in 2023 into MNSI (Maternity and Newborn Safety Investigations) and HSSIB (Health Services Safety Investigations Body)

Produced reports with investigation findings and recommendations



Health Services Safety
Investigations Body

Maternity safety investigations



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Factors affecting the delivery of safe care in midwifery units

8 May 2024 National learning report

National learning report.

[Read Publication](#)

Learning from maternal death investigations during the first wave of the COVID-19 pandemic

25 February 2021 National learning report HSIB legacy publication

National learning report.

[Read Publication](#)

East Kent Hospitals University NHS Foundation Trust: HSIB summary report

6 April 2020 HSIB legacy publication

A summary of the investigations undertaken with East Kent Hospitals University NHS Foundation Trust.

[Read Publication](#)

Assessment of risk during the maternity pathway

2 March 2023 National learning report HSIB legacy publication

National learning report.

[Read Publication](#)

National learning report

This report is intended for healthcare organisations, policymakers and the public. It is based on a thematic analysis (a process that looks for common themes) of 92 maternity investigation reports, where the safety incident under investigation included care provided in a midwifery unit.

Midwifery units are staffed by midwives and support staff. Typically, pregnant women who choose to give birth in a midwifery unit have been assessed as having a low chance of complications during labour and birth. Sometimes a pregnant woman or baby may need to be transferred from a midwifery unit to an obstetric unit (a hospital unit where specialist doctors are primarily responsible for their care) to receive additional care and treatment.

The thematic analysis identified 4 main themes and findings, which include issues relating to:

1. **Work demands and capacity to respond** – the number of tasks needed to be done and whether there are enough (and suitable) staff, and appropriate physical space, to do them.
2. **Intermittent auscultation** – a method used to assess a baby's heart rate as an indicator of their wellbeing.
3. **How prepared an organisation is for predictable safety-critical scenarios**, and the role played by in situ simulation (a training method that involves staff rehearsing scenarios in the workplace).
4. **Telephone triage** – the assessment a midwife carries out when a pregnant woman telephones because they have gone into labour or have a concern about their pregnancy.

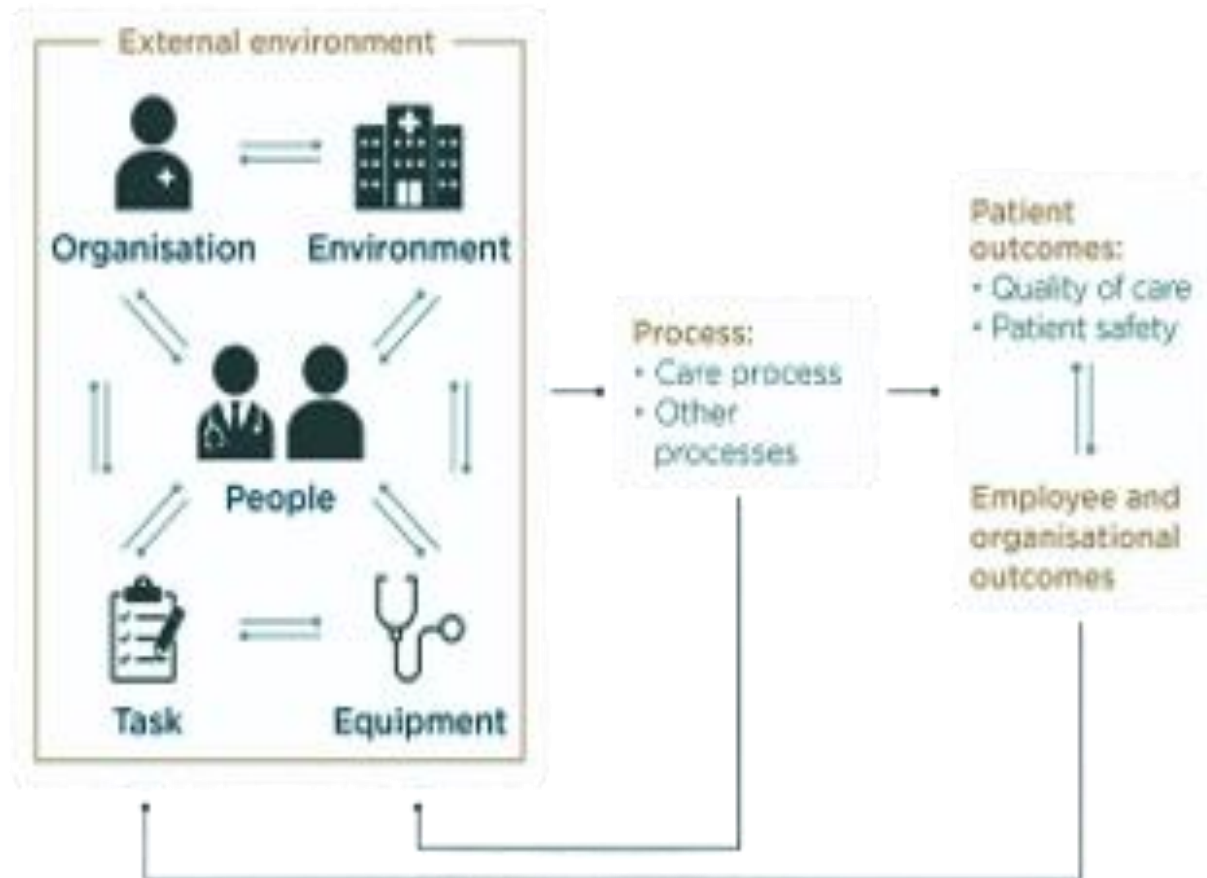


To develop a reliable and unbiased machine learning based tool to extract and analyse intelligence from maternity investigation reports using human factor concepts specifically designed for maternity investigations.

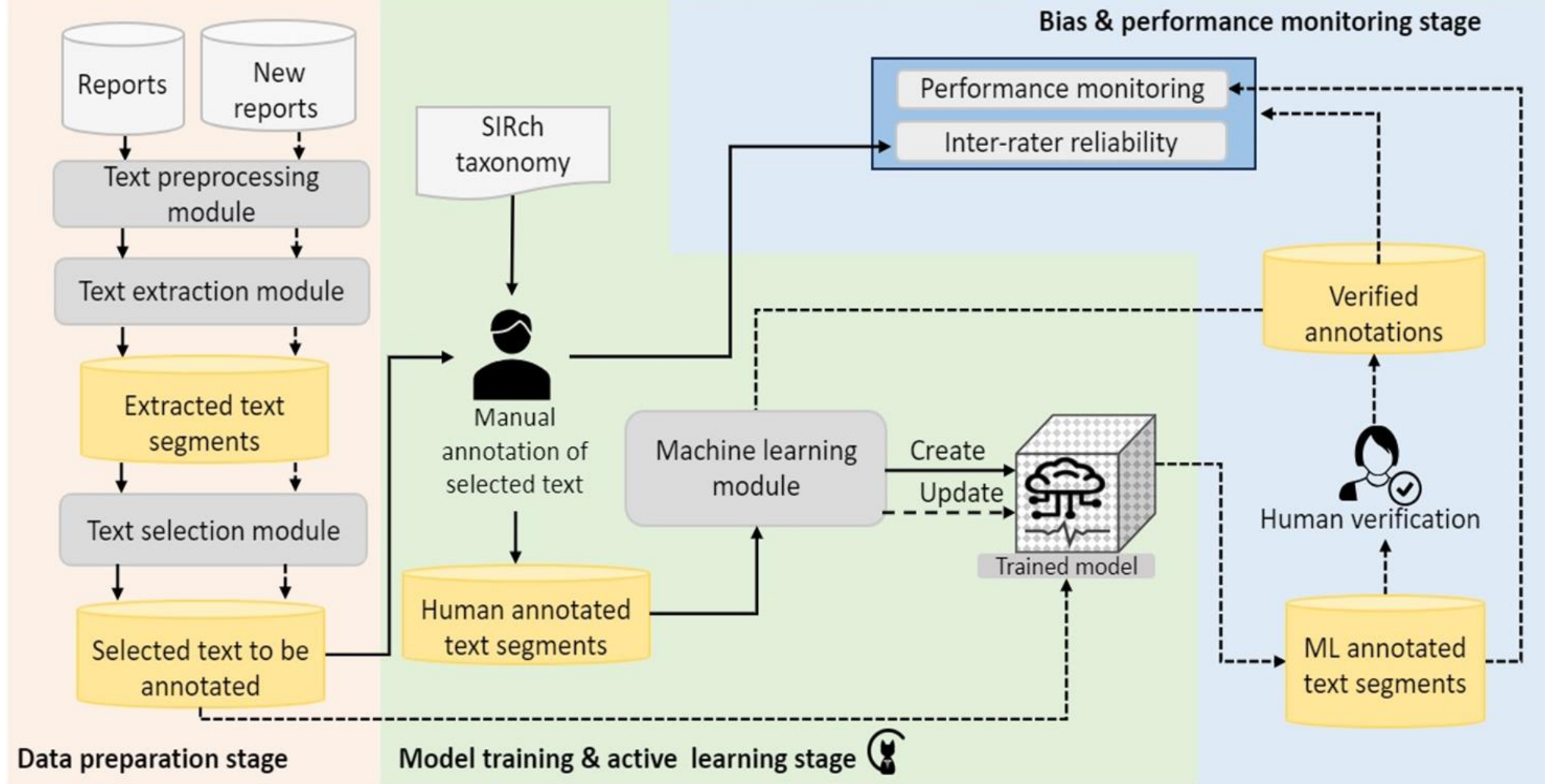
- Learn to map each sentence to a human factors concept from the taxonomy
- Automatically annotate reports and present an analysis of the concepts
- Unbiased towards specific ethnicity/factors

I-SIRCH extracting intelligence from investigation reports & beyond

Supplementary File 1: SIRch taxonomy of human factors for maternity investigations



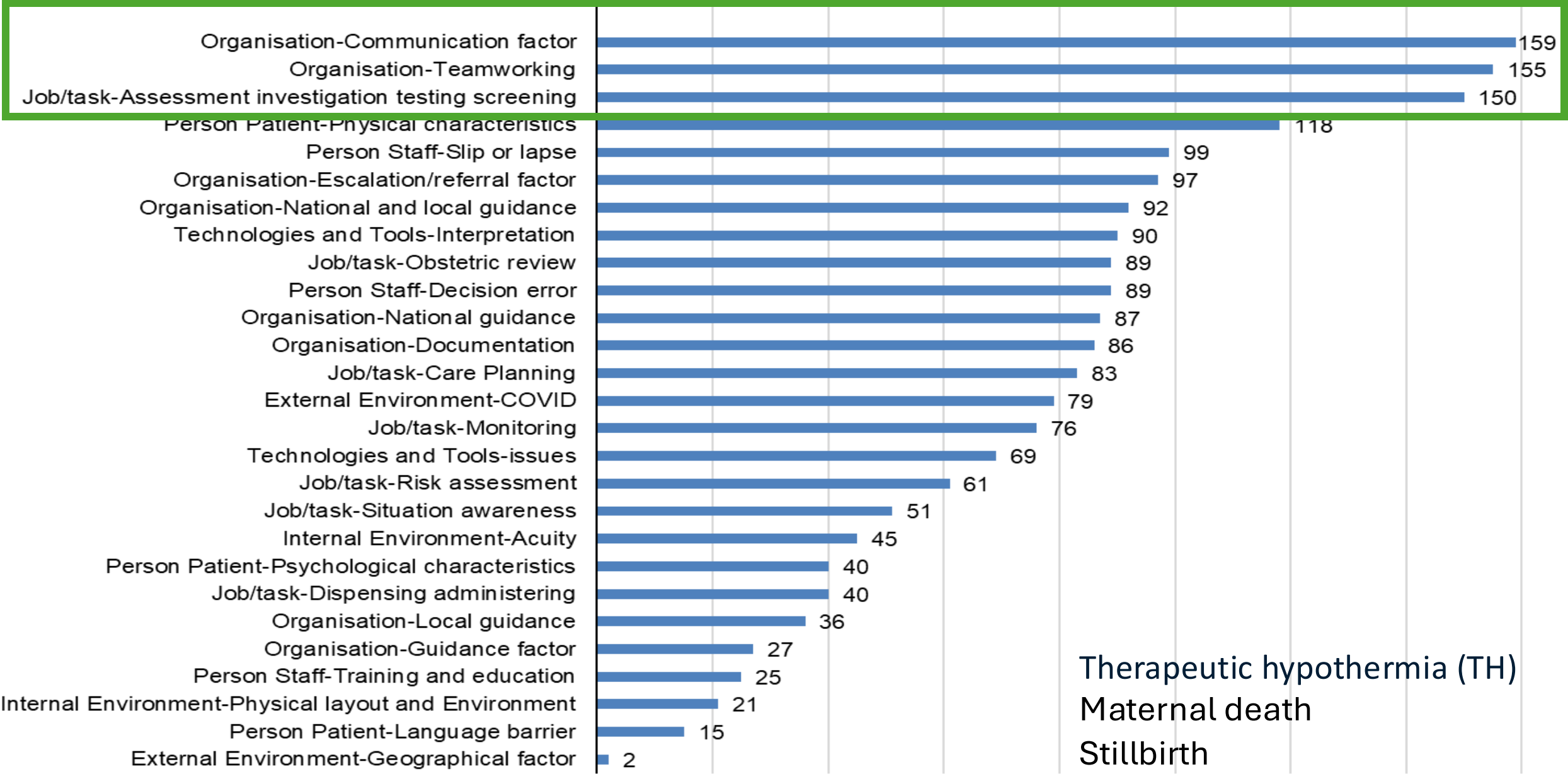
- | | |
|---|--|
| <ol style="list-style-type: none"> 1. External Environment <ol style="list-style-type: none"> a. Policy factor b. Societal factor c. Economic factor d. COVID e. Geographical factor (e.g. Location of patient) 2. Internal Environment <ol style="list-style-type: none"> a. Physical layout and Environment b. Acuity (e.g. capacity of the maternity unit as a whole) c. Availability (e.g. operating theatres) d. Time of day (e.g. night working or day of the week) 3. Organisation <ol style="list-style-type: none"> a. Team culture factor (e.g. patient safety culture) b. Incentive factor (e.g. performance evaluation) c. Teamworking d. Communication factor <ol style="list-style-type: none"> i. Between staff ii. Between staff and patient (verbal) e. Documentation f. Escalation/referral factor (including fresh eyes reviews) g. National and/or local guidance h. Language barrier 4. Jobs/Task <ol style="list-style-type: none"> a. Assessment, investigation, testing, screening (e.g. holistic review) b. Care planning c. Dispensing, administering d. Monitoring e. Risk assessment f. Situation awareness (e.g. loss of helicopter view) g. Obstetric review 5. Technologies and Tools <ol style="list-style-type: none"> a. Issues b. Interpretation (e.g. CTG) | <ol style="list-style-type: none"> 6. Person <ol style="list-style-type: none"> a. Patient (characteristics and performance) <ol style="list-style-type: none"> i. Characteristics <ol style="list-style-type: none"> A. Physical characteristics B. Psychological characteristics (e.g. stress, mental health) C. Language competence (English) D. Disability (e.g. hearing problems) E. Training and education (e.g. attendance at antenatal classes) F. Record of attendance (e.g. failure to attend antenatal classes) ii Performance <ol style="list-style-type: none"> A. Slip or lapse (errors that tend to happen in routine tasks that people are doing without much conscious thought) B. Decision error (errors in conscious judgements, decisions due to lack of knowledge and from misunderstanding of a situation) C. Intentional rule breaking (deliberately do something different from rules) b. Staff (characteristics and performance) <ol style="list-style-type: none"> i. Characteristics <ol style="list-style-type: none"> A. Physical characteristics B. Psychological characteristics (e.g. stress, mental health) C. Language competence (English) D. Disability (e.g. hearing problems) E. Training and education (e.g. attendance at ante-natal classes) F. Record of attendance (e.g. failure to attend antenatal classes) ii Performance <ol style="list-style-type: none"> A. Slip or lapse (errors that tend to happen in routine tasks that people are doing without much conscious thought) B. Decision error (errors in conscious judgements, decisions due to lack of knowledge and from misunderstanding of a situation) C. Intentional rule breaking (deliberately do something different from rules) |
|---|--|



The model achieved a high level of accuracy, correctly identifying relevant concepts in 90% of the sentences from 97 reports.

Applying I-SIRch to analyse these reports revealed that certain human factors disproportionately affected mothers from different ethnic groups.

In particular, gaps in risk assessment were more prevalent for minority mothers, whilst communication issues were common across all groups but potentially more for minorities.



Therapeutic hypothermia (TH)
Maternal death
Stillbirth

188 reports, 3122 sentences

Number of reports per concept

Communication

Not listening to, dismissing or ignoring women's concerns, preferences and requests

Not providing sufficient information for informed decisions

Conflicting recollections of discussions

Problematic staff interactions with women and their families



Communication

Lack of translation services, leading to concerns not being understood and lack of discussions around options, risks and care

Information provided, and confirmations around health history, risks in pregnancy, signs of infection, and post-discharge monitoring needed



Communication

Calls not being answered, lack of communication around urgency pre-hospital and in hospital

Delays informing family leading to less time with baby before death



Communication

Disputes around language used regarding baby's condition

Lack of debrief including all staff involved

Missed opportunities to communicate around risks and abnormal results that would have informed decision-making and care pathway changes





Teamworking

Fragmented triage care with many staff and inconsistent handovers

Communication and handover issues between teams

Recurring handover issues, not sharing key details like risk factors and urgent decisions



Teamworking

Confusing guidance leading to role uncertainties, delays expediting care

Hierarchy perceptions and communication gaps on needed interventions

Ambulance delays. Excess workloads without adequate cover



Assessment, investigation, testing, screening (e.g., holistic review)

Gaps in assessments and documentation and lack of key lab tests represented

Failure to recognise start of second stage of labour

Failures to appreciate full clinical pictures, perform key examinations, and meet standards

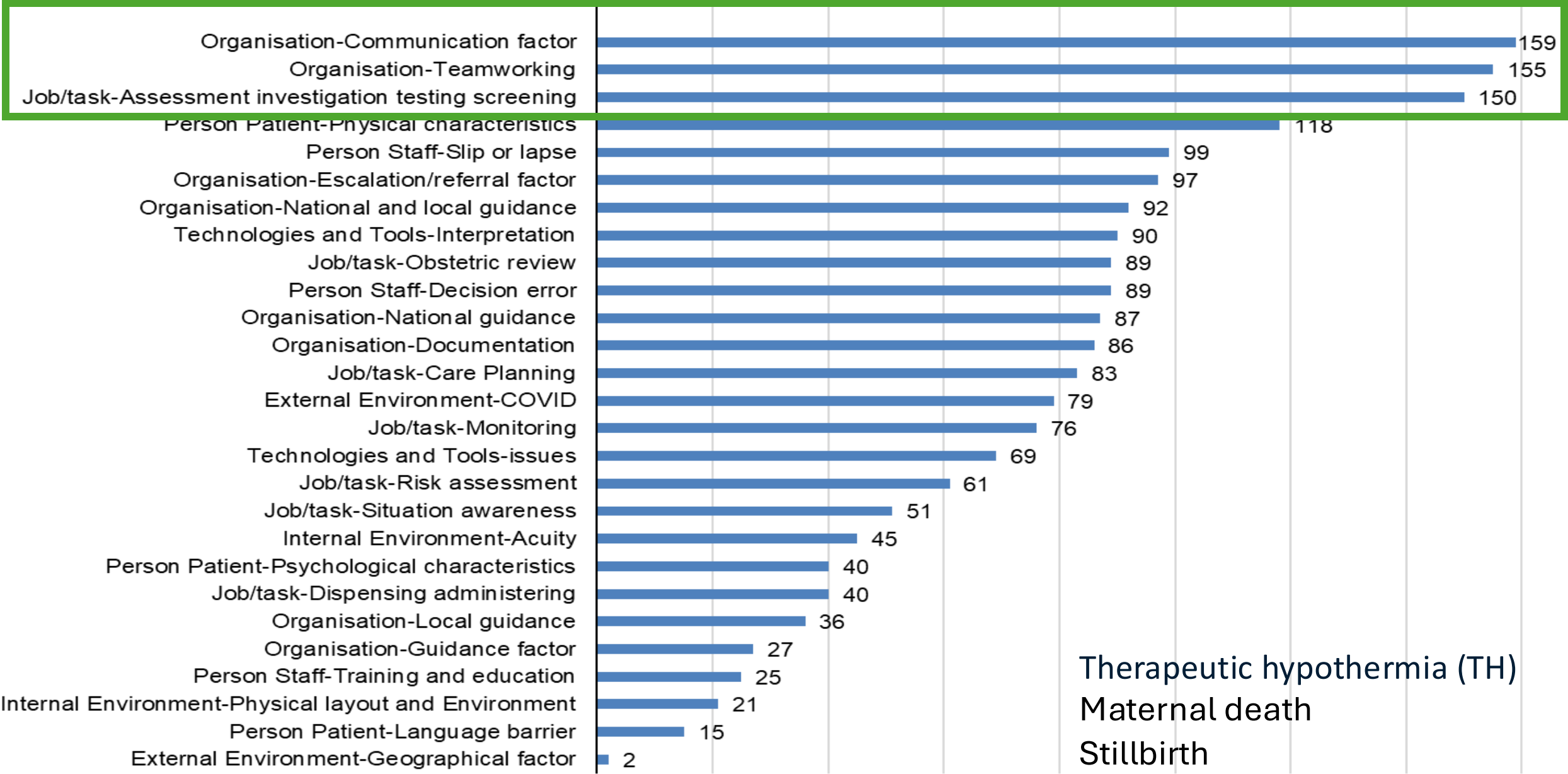


Assessment, investigation, testing, screening (e.g., holistic review)

Inadequate clinical assessments and lack of holistic reviews during mothers' care

Missed assessments, examinations, and reviews

Busy units prevented appreciating mothers' concerns and performing needed assessments

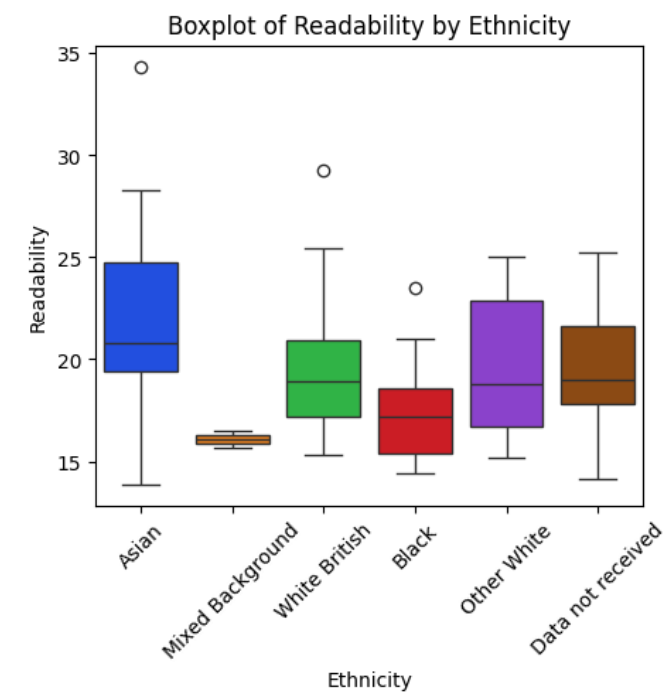
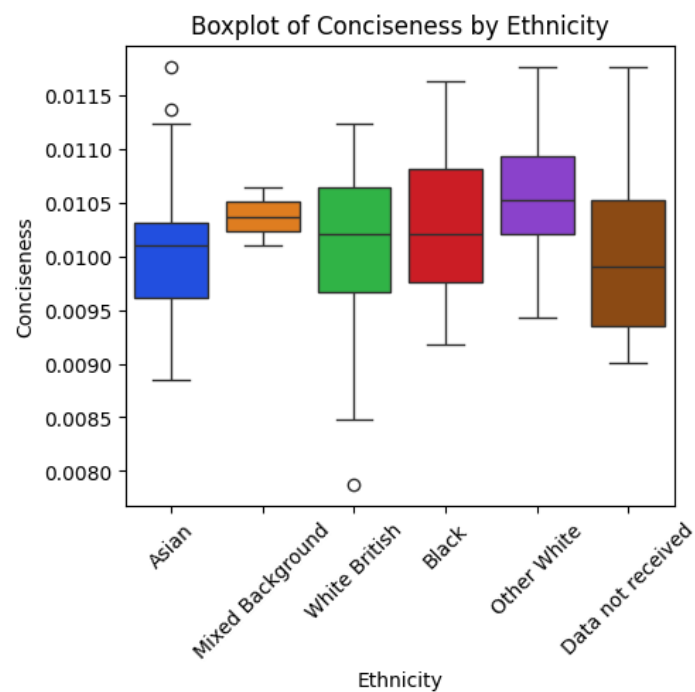
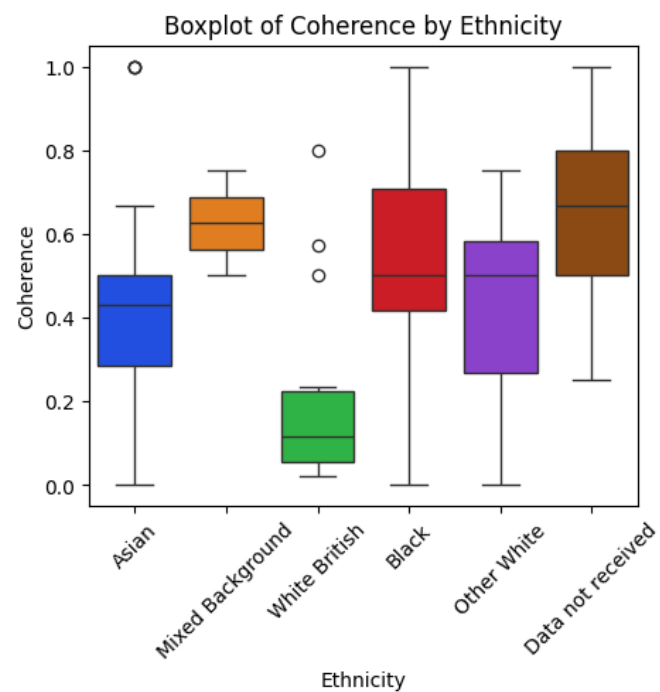
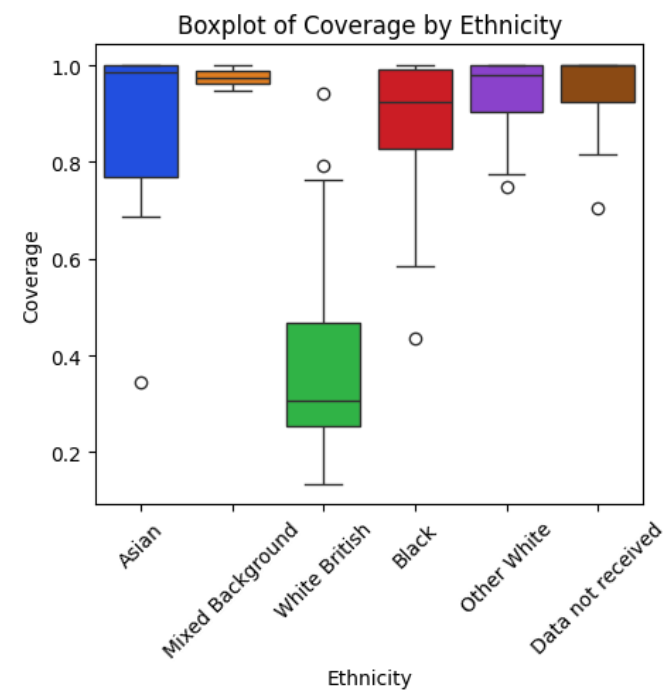
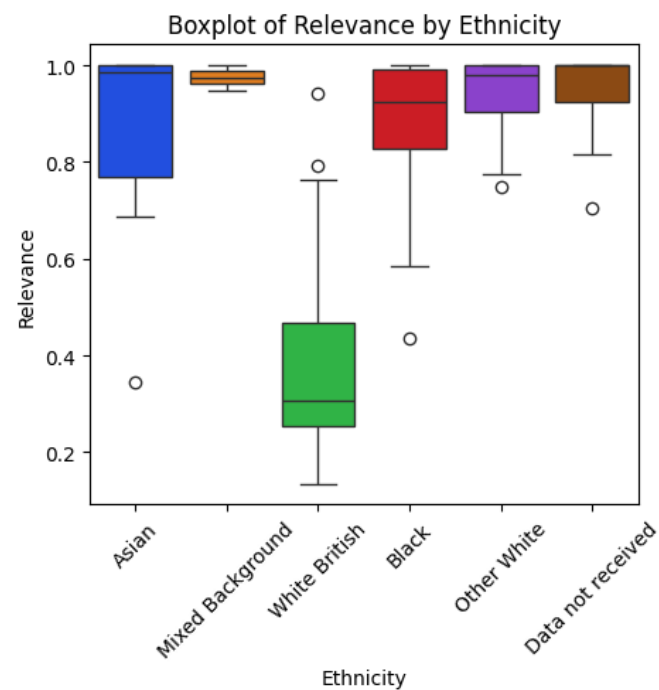
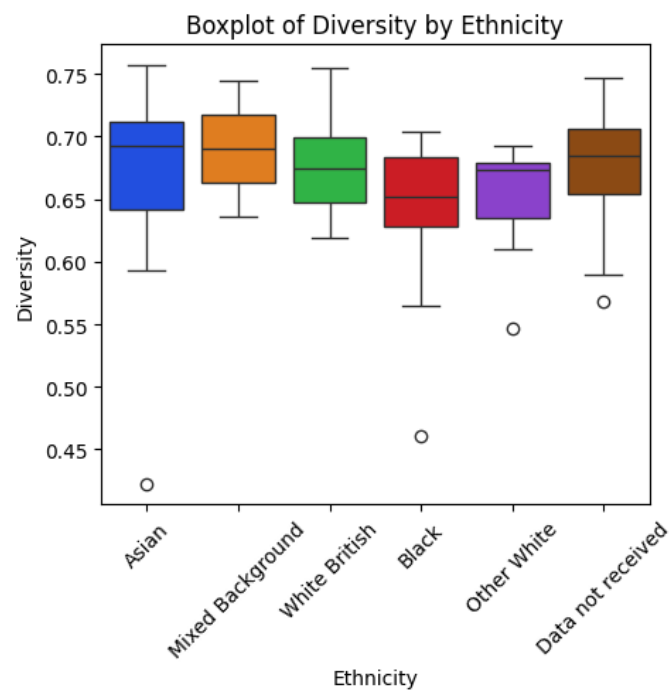


Therapeutic hypothermia (TH)
Maternal death
Stillbirth

188 reports, 3122 sentences

Number of reports per concept

Ethnic Group	Outcome	Summary	Related IDs	Sentence
Asian	Cooled baby	High workload prevented timely review of abnormal CTGs and recognition of risk factors, delaying intervention and impacting cooled baby outcomes.	370, 373, 374	
Asian	Early NND	Delays in post-delivery assessments of a baby prevented recognition of condition and need for monitoring, impacting an early NND.		372
Asian	Stillbirth	Incomplete maternal observations and assessments contributed to lack of recognition of clinical deterioration prior to stillbirths.	366, 367, 368	
Black	Cooled baby	Lack of holistic assessments meant missed screening for gestational diabetes and growth scans, potentially impacting cooled baby outcomes.	289, 290, 291, 292	
Black	MD	Lack of pregnancy test and referral to additional support represented missed care opportunities prior to a maternal death.		377, 378
Data not received	MD	Gaps in assessments and documentation, and lack of key lab tests represented missed opportunities prior to maternal deaths.	379, 380, 381	
Mixed Background	Cooled baby	Lack of testing for Zika virus prevented full risk assessment for a cooled baby.		371



Artificial Intelligence in Healthcare



- **Annotation:** Natural Language Processing (NLP) & Machine learning to extract intelligence from reports and annotate them (with human in the loop)
- Offline generative text **summarisation** tool (summarises ethnicity, factors, & incident)
- **Topic modelling** to identify themes in the text (ethnicity & incident)
- **Robust evaluations** during development to ensure performance is fair and not biased towards certain groups

Publications:

- Singh, MK, Cosma, G, Waterson, P, Back, J, Jun, GT (2024) I-SIRch: AI-Powered Concept Annotation Tool For Equitable Extraction And Analysis Of Safety Insights From Maternity Investigations, Accepted. *To Appear in* International Journal of Population Data Science.
- Cosma, G, Singh, MK, Waterson, P, Jun, GT, Back, J (2024) [Unveiling disparities in maternity care: a topic modelling approach to analysing maternity incident investigation reports](#). Lecture Notes in Computer Science, vol 14975. Springer, Cham. https://doi.org/10.1007/978-3-031-67278-1_23
- Cosma, G., Singh, M.K., Waterson, P., Jun, G.T., Back, J. (2024). Intelligent Multi-document Summarisation for Extracting Insights on Racial Inequalities from Maternity Incident Investigation Reports. Lecture Notes in Computer Science, vol 14976. Springer, Cham. https://doi.org/10.1007/978-3-031-67285-9_23

What do Prevention of Future Death reports reveal about the key safety themes and system-level failures in UK maternity care, and how can automated multi-framework analysis enhance patient safety intelligence?

LENS.ai - Learning from Evidence through Natural language Systems

What are PFD Reports? Independent judicial assessments issued by coroners following inquests into unexpected deaths, providing cross-organisational perspectives on patient safety issues.

UK Judiciary PFD Reports Analysis

This application analyses Prevention of Future Deaths (PFD) reports from the UK Judiciary website to uncover patterns, themes, and insights.

How to Use This Tool

(1) 🔍 Scrape Reports

(2) 📁 Scraped File Preparation

(3) 📊 Scraped File Analysis

(4) 📄 Topic Analysis & Summaries

(5) 📝 Concept Annotation

(6) 📊 Theme Analysis Dashboard

Search tool for Prevention of Future Deaths (PFD) reports from the UK Judiciary website.

- Extract detailed PFD reports with metadata, full content, and associated PDFs
- Filtering by keywords, categories, and date ranges
- Export options in CSV and Excel formats

Handling Large Result Sets: For extensive search results, use the 'Start page' and 'End page' number inputs to download reports in manageable batches.

Scrape PFD Reports

Search keywords:

PFD Report type:

Select element

Sort by:

Relevance

Filter search

Published on or after

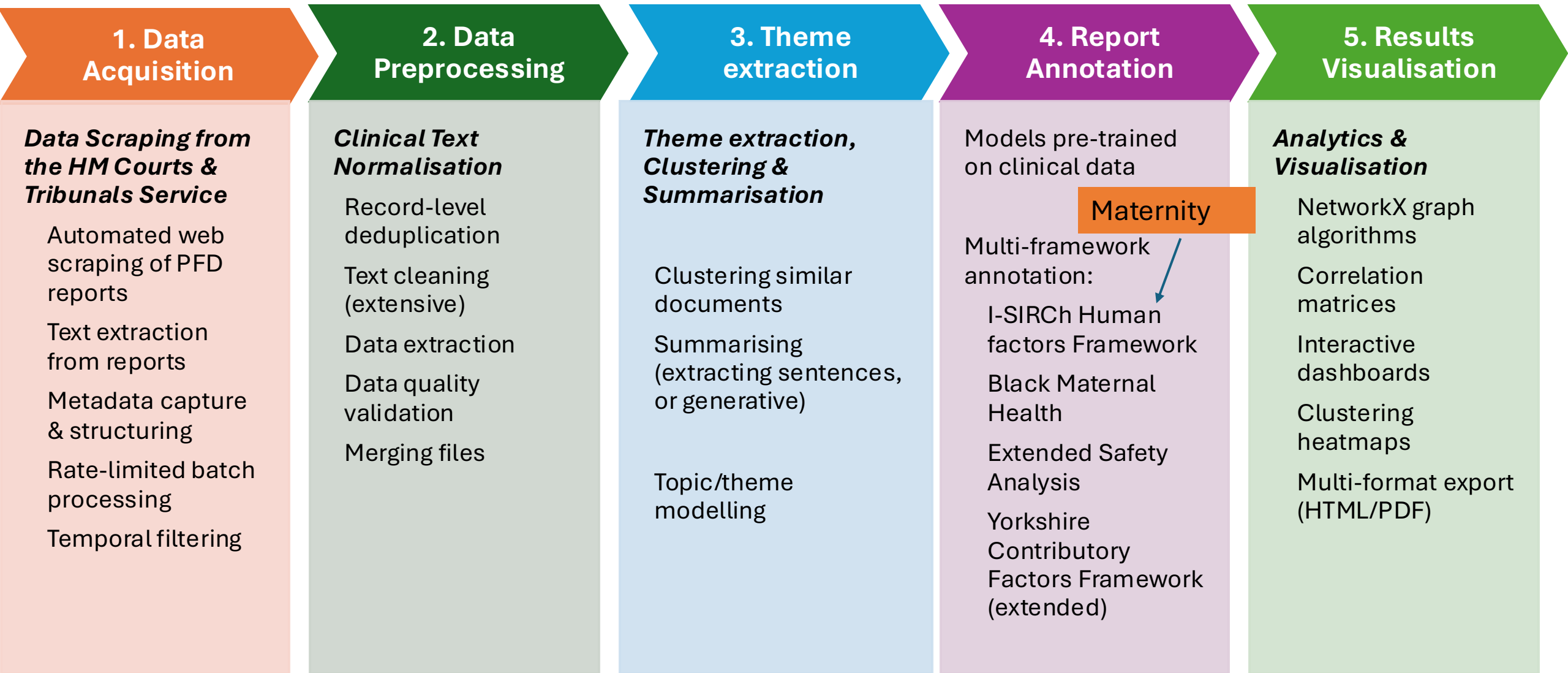
For example, 27 3 2007

Day

Month

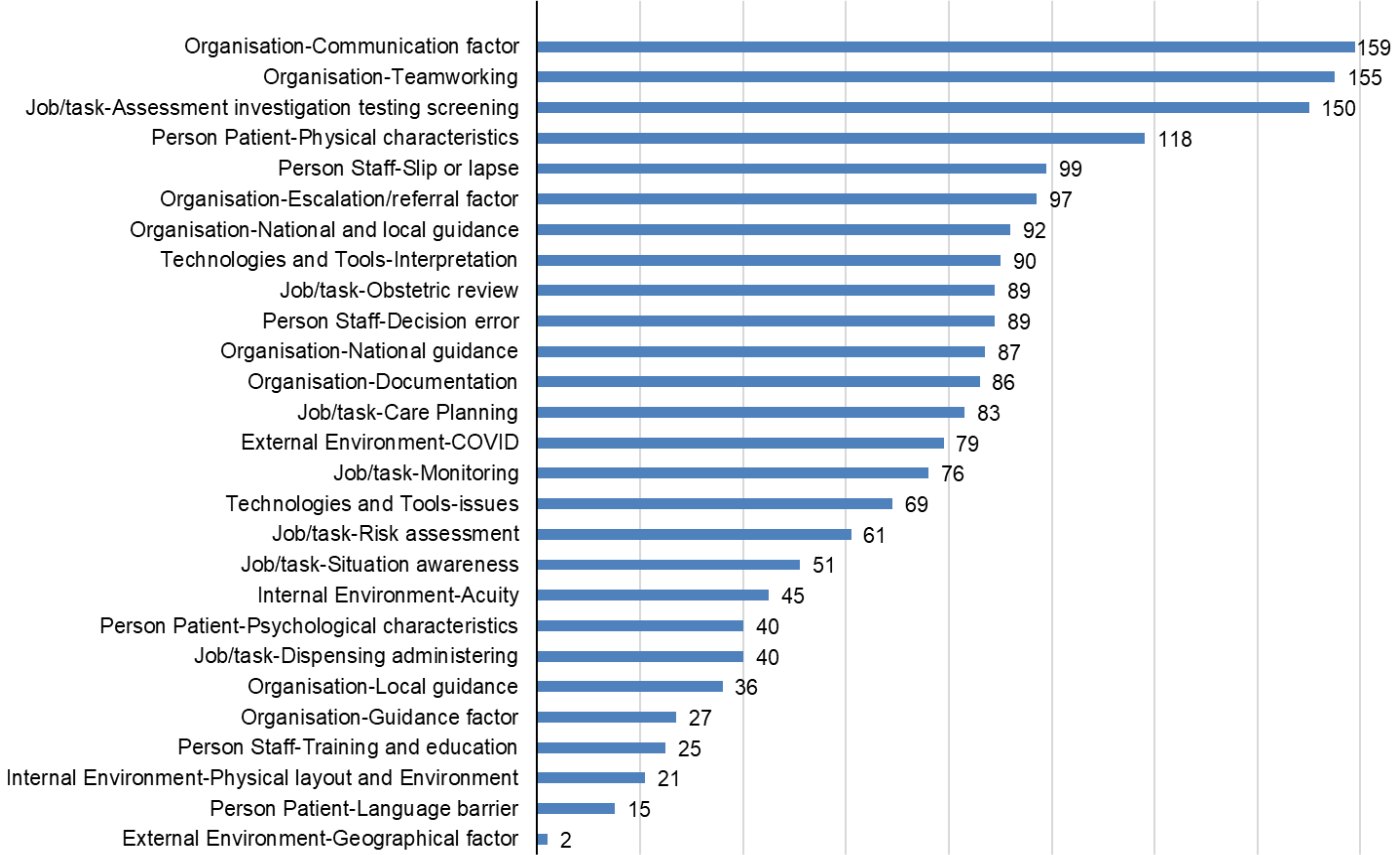
Year

I-SIRCH extracting intelligence from investigation reports & beyond



HSIB-I-SIRch framework only

PFD reports (n=51) (using 3 frameworks[+,-,*])	
Medication Safety (*)	78.4%
Staff Performance (+)	78.4%
Team Culture (+)	74.5%
Informed consent/agency (-)	66.7%
Diagnostic Testing & Specimens (*)	64.7%
Patient Record Attendance (+)	62.7%
Physical Layout & Environment (+)	56.9%
Staff Decision Error (+)	41.2%
Peer Support & Supervision (*)	39.2%
External Societal Factor (+)	39.2%



House of Commons Women & Equalities Committee themes in PFD reports:
informed consent/agency ranked 4th (66.7%), followed by guidance gaps (37.3%),
care quality issues (31.4%), and communication (29.4%).

I-SIRCH extracting intelligence from investigation reports & beyond

1. Concept annotation

[illegible]

Document: Eclipse Morrison: Prevention of future deaths report

Identified Themes

Framework	Theme	Confidence	Score	Matched Keywords	Color
Extended Analysis	Diagnostic Testing and Specimen Handling	High	0.852	analysis, interpretation	☐
House of Commons	Guidance gaps	High	0.886	information, guidance	☐
House of Commons	Fragmented care	High	0.883	clear, care	☐
House of Commons	Care quality and access issues	High	0.881	access, quality, treatment	☐
House of Commons	Inadequate maternity care	High	0.868	maternity, care	☐
House of Commons	Socioeconomic factors and deprivation	High	0.863	factors	☐
I-SIRch	Organisation - National and/or local guidance	High	0.884	guidance	☐
I-SIRch	Person - Patient (characteristics and performance) - Characteristics - Training and education (e.g., attendance at ante-natal classes)	High	0.876	training, education	☐
I-SIRch	Jobs/Task - Care planning	High	0.872	care, planning	☐
I-SIRch	Person - Patient (characteristics and performance) - Characteristics - Record of attendance (e.g., failure to attend antenatal classes)	High	0.865	record	☐
I-SIRch	External - Geographical factor (e.g. Location of patient)	High	0.86	factor	☐

Highlighted Text

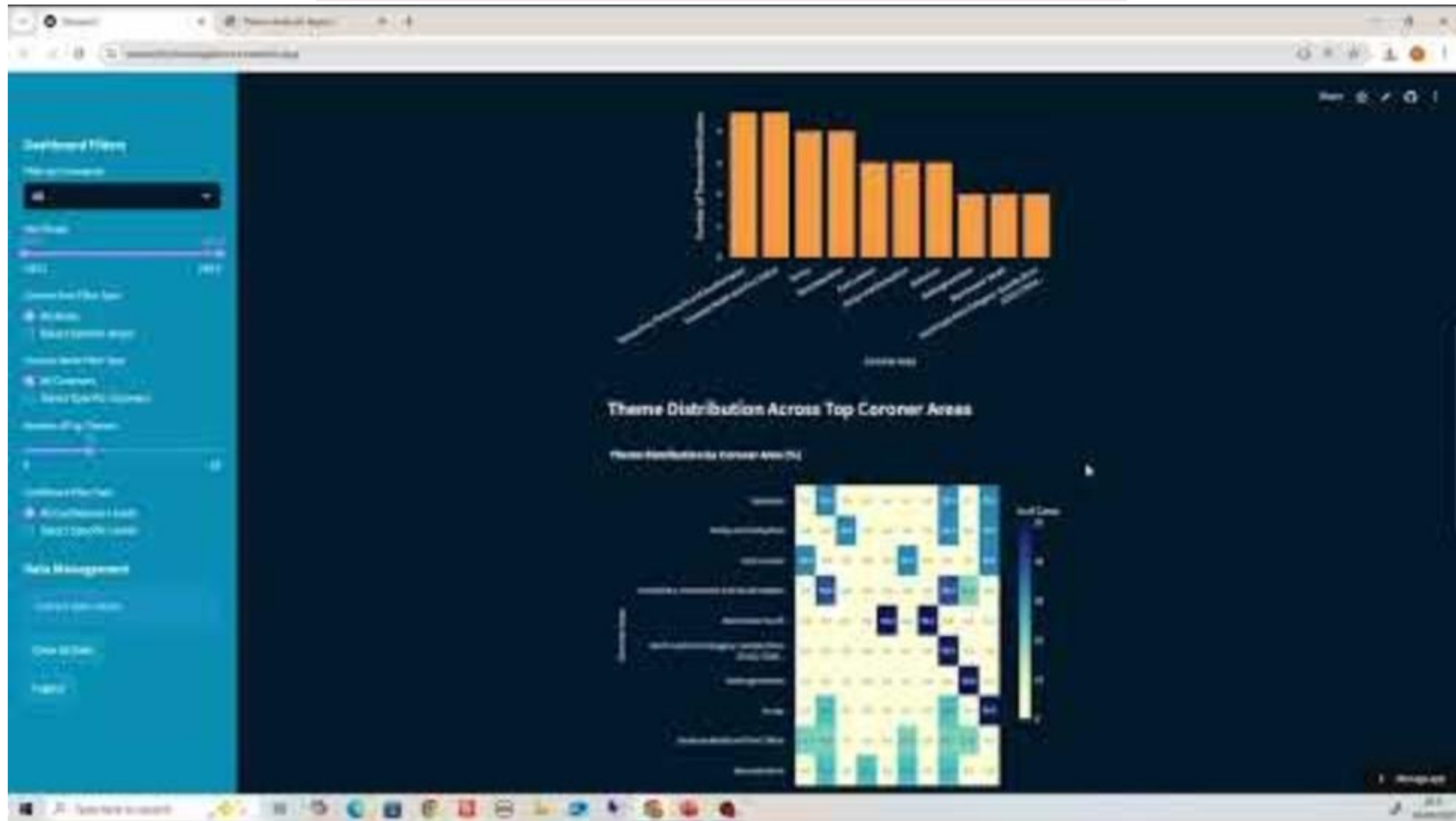
are as follows - Evidence was given at the inquest and it is accepted, that many of the concerns identified in relation to Eclipse's birth have now been addressed, but some remain. 1. Risk analysis / mode and timing of birth A significant factor was the failure to consider an elective Caesarean Section (CS). It is not clear why on this history and consideration of mode of delivery did not include a consideration of an elective CS at any stage, by any of the doctors involved. Evidence was given that existing policies in place at GEH were not followed and the reason for this failure has not been identified. Whilst Badgernet could be a useful tool, the evidence suggested that it is being regarded as a first line of defence rather than a failsafe. I am told that the Badgernet maternity system implemented in October 2021 will resolve the problems relating to mode and timing of birth, in that the system will require a consultant to approve a decision for CS/induction of labour (IOL) and that IOL's will not be approved by the labour ward coordinator unless the paper booking has been approved on paper. However, on the evidence received it is not clear how this will assist if the risk factors and the need to consider IOL/elective CS have not been identified, by junior staff, thus triggering the need for escalation to/approval by a consultant. I am not clear what has been done to ensure that junior doctors and locums have sufficient technical knowledge to ensure that they are able to identify serious risk factors and alert the consultant to these, so that the consultant may consider the appropriate mode of delivery. No information was provided as to the availability of regular face to face training for all grades concerning high-risk pregnancies not just for career trainees and foundation doctors. I am told that a memo was sent to all junior doctors reminding them that any plan for either IOL or elective CS must be approved by a consultant and that an induction pack containing that information is provided to new starters and locums. I have seen the Women's and Children Clinical Education Guideline introduced in November 2022. I am told that the information was placed in a prominent position on notice boards, staff rooms and in blogs. It is not clear how the assimilation of this knowledge is tested. It was suggested that this may be in appraisal, but this seems only likely to identify problems after they arise. In any event it seems these methods of dissemination of information and the appraisal system were in place at the time of Eclipse's birth but did not ensure that the doctors involved in her mother's care appreciated the impact of the risk factors in this pregnancy. It was not explained how Badgernet or any policy or procedure in place, would ensure that there is a holistic review (including ultrasound scan findings) when planning for timing and mode of birth. The concern remains that there will be further failure to ensure that appropriate timing and mode of delivery will be provided in high-risk pregnancies. 2. Access to Badgernet portal / full implementation of the Badgernet software It seems that the Badgernet system is being relied on to address a number of issues which were identified in this case, and heavy reliance is being placed on a system which is not yet fully implemented. The concern remains that a critical aspect of this system, access to the portal, will not be in place until autumn of 2023 at the earliest. 3. Quality Assessment I am informed that Badgernet can easily identify fields which have not been completed and will prevent a record being closed until the field is completed, but it cannot identify the quality of any such entries. I am told that the quality checks are made on ten sets of notes per month out of an estimated 3,000 records that will be open at any one time. The concern remains that there is insufficient quality assurance in this system. 4. Procedure for escalating concerns arising out of Ultrasound Scans (USS) I understand that the procedure for escalating concerns arising out of a USS where it is not possible to obtain an accurate measurement in a high-risk pregnancy is currently under review. I am told that the new policy is not in place. The concern remains that no date has been set for its implementation. 5. Counselling for mothers at increased risk of shoulder dystocia I have not seen any indication that all mothers identified to have an increased chance of shoulder dystocia now receive counselling regarding the risks and benefits associated with vaginal birth or CS. Assisting mothers to understand the implications of risks they face is fundamental to supporting them to make fully informed decisions, in accordance with Montgomery. The concern remains that there is no clear plan in place to ensure mothers receive the support they require to make fully informed decisions in relation to mode of delivery where there is risk of shoulder dystocia. 6. Interpretation of Montgomery Ante-natal care is unique in that decisions have to be made for the benefit of two patients (the mother and the baby) and the treatment options for each may, as in this case have competing risks and benefits. In addition, in ante-natal care, the circumstances may change and action may need to be taken very quickly. On the basis of evidence given at the inquest, there seems to be a lack of clarity as to the way in which Montgomery guidelines are interpreted. It was acknowledged in evidence that parents often want a steer as to the best/safest course of action and that may require medical professionals to express opinions as to the weight to be placed on different risk factors. In some cases, parents may prefer to rely on the viewpoint of an experienced medical professional. It seems that medical professionals do not feel they can offer this assistance as it might be interpreted as trying to impose their opinion on the parent. The way in which Montgomery is interpreted and the extent to which medical professionals can offer an opinion is of wider concern than just the actions of those at GEH and should be considered by those who produce the guidance and deliver training to medical professionals.

Framework	Theme	Color	Matched Keywords	Extracted Sentence
Extended Analysis	Diagnostic Testing and Specimen Handling + I-SIRch_External - Geographical factor (e.g. Location of patient)	#D1ECF1	analysis + factor	Risk analysis / mode and timing of birth A significant factor was the failure to consider an elective Caesarean Section (CS).
House of Commons	Fragmented care	#FFECB3	clear	It is not clear why on this history and consideration of mode of delivery did not include a consideration of an elective CS at any stage, by any of the doctors involved.
House of Commons	Inadequate maternity care	#F3E5F5	maternity	I am told that the Badgernet maternity system implemented in October 2021 will resolve the problems relating to mode and timing of birth, in that the system will require a consultant to approve a decision for CS/induction of labour (IOL) and that IOL's will not be approved by the labour ward coordinator unless the paper booking has been approved on paper.

increased risk of shoulder dystocia I have not seen any indication that all mothers identified to have an increased chance of shoulder dystocia now receive counselling regarding the risks and benefits associated with vaginal birth or CS. Assisting mothers to understand the implications of risks they face is fundamental to supporting them to make fully informed decisions, in accordance with Montgomery. The concern remains that there is no clear plan in place to ensure mothers receive the support they require to make fully informed decisions in relation to mode of delivery where there is risk of shoulder dystocia. 6. Interpretation of Montgomery Ante-natal care is unique in that decisions have to be made for the benefit of two patients (the mother and the baby) and the treatment options for each may, as in this case have competing risks and benefits. In addition, in ante-natal care, the circumstances may change and action may need to be taken very quickly. On the basis of evidence given at the inquest, there seems to be a lack of clarity as to the way in which Montgomery guidelines are interpreted. It was acknowledged in evidence that parents often want a steer as to the best/safest course of action and that may require medical professionals to express opinions as to the weight to be placed on different risk factors. In some cases, parents may prefer to rely on the viewpoint of an experienced medical professional. It seems that medical professionals do not feel they can offer this assistance as it might be interpreted as trying to impose their opinion on the parent. The way in which Montgomery is interpreted and the extent to which medical professionals can offer an opinion is of wider concern than just the actions of those at GEH and should be considered by those who produce the guidance and deliver training to medical professionals.

Framework	Theme	Color	Matched Keywords	Extracted Sentence
Extended Analysis	Diagnostic Testing and Specimen Handling + I-SIRch_External - Geographical factor (e.g. Location of patient)	#D1ECF1	analysis + factor	Risk analysis / mode and timing of birth A significant factor was the failure to consider an elective Caesarean Section (CS).
House of Commons	Fragmented care	#FFECB3	clear	It is not clear why on this history and consideration of mode of delivery did not include a consideration of an elective CS at any stage, by any of the doctors involved.
House of Commons	Inadequate maternity care	#F3E5F5	maternity	I am told that the Badgernet maternity system implemented in October 2021 will resolve the problems relating to mode and timing of birth, in that the system will require a consultant to approve a decision for CS/induction of labour (IOL) and that IOL's will not be approved by the labour ward coordinator unless the paper booking has been approved on paper.
House of Commons	Fragmented care + House of Commons_Socioeconomic factors and deprivation	#FFECB3	clear + factors	However, on the evidence received it is not clear how this will assist if the risk factors and the need to consider IOL/elective CS have not been identified, by junior staff, thus triggering the need for escalation to/approval by a consultant.
House of Commons	Fragmented care + House of Commons_Socioeconomic factors and deprivation	#FFECB3	clear + factors	I am not clear what has been done to ensure that junior doctors and locums have sufficient technical knowledge to ensure that they are able to identify serious risk factors and alert the consultant to these, so that the consultant may consider the appropriate mode of delivery.
House of Commons	Guidance gaps + I-SIRch_Person - Patient (characteristics and performance) - Characteristics - Training and education (e.g., attendance at ante-natal classes)	#E1F5FE	information + training	No information was provided as to the availability of regular face to face training for all grades concerning high-risk pregnancies not just for career trainees and foundation doctors.
House of Commons	Guidance gaps	#E1F5FE	information	I am told that a memo was sent to all junior doctors reminding them that any plan for either IOL or elective CS must be approved by a consultant and that an induction pack containing that information is provided to new starters and locums.
I-SIRch	Person - Patient (characteristics and performance) - Characteristics - Training and education (e.g., attendance at ante-natal classes)	#D1E7DD	education	I have seen the Women's and Children Clinical Education Guideline introduced in November 2022.
House of Commons	Guidance gaps	#E1F5FE	information	I am told that the information was placed in a prominent position on notice boards, staff rooms and in blogs.
House of Commons	Fragmented care	#FFECB3	clear	It is not clear how the assimilation of this knowledge is tested.
House of Commons	Fragmented care + House of Commons_Guidance gaps + House of Commons_Inadequate maternity care + House of Commons_Socioeconomic factors and deprivation + I-SIRch_Jobs/Task - Care planning	#FFECB3	care + information + care + factors + care	In any event it seems these methods of dissemination of information and the appraisal system were in place at the time of Eclipse's birth but did not ensure that the doctors involved in her mother's care appreciated the impact of the risk factors in this pregnancy.
I-SIRch	Jobs/Task - Care planning	#FFD580	planning	It was not explained how Badgernet or any policy or procedure in place, would ensure that there is a holistic review (including ultrasound scan findings) when planning for timing and mode of birth.
House of Commons	Care quality and access issues	#FFF3E0	access	Access to Badgernet portal / full implementation of the Badgernet software It seems that the Badgernet system is being relied on to address a number of issues which were identified in this case, and heavy reliance is being placed on a system which is not yet fully implemented.
House of				

2. Analysis

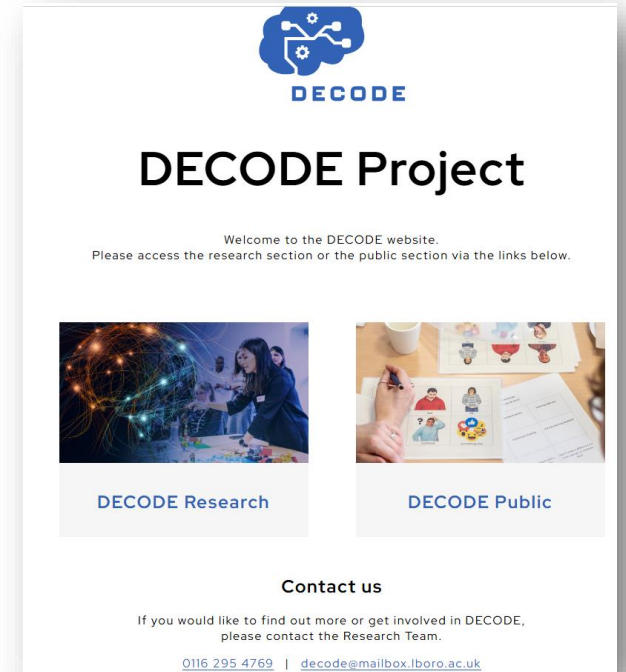


DECODE

Data-driven machine-learning aided stratification and management of multiple long-term Conditions in adults with intellectual disabilities (DECODE)

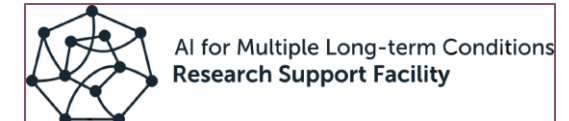


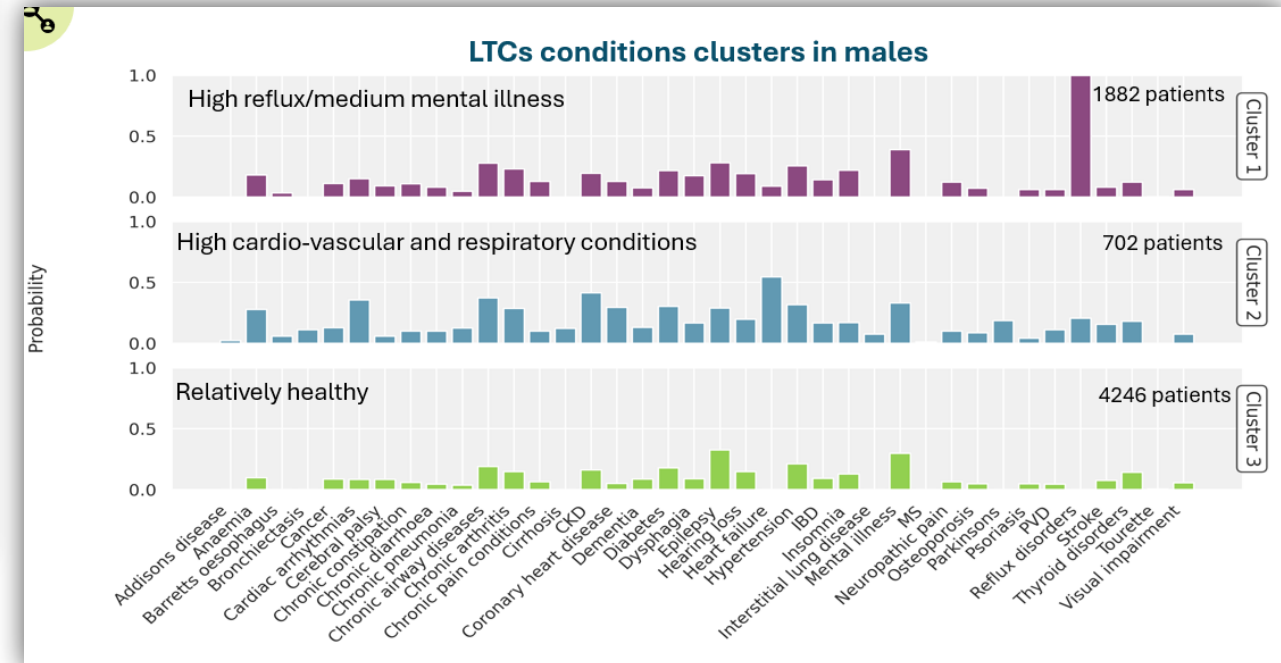
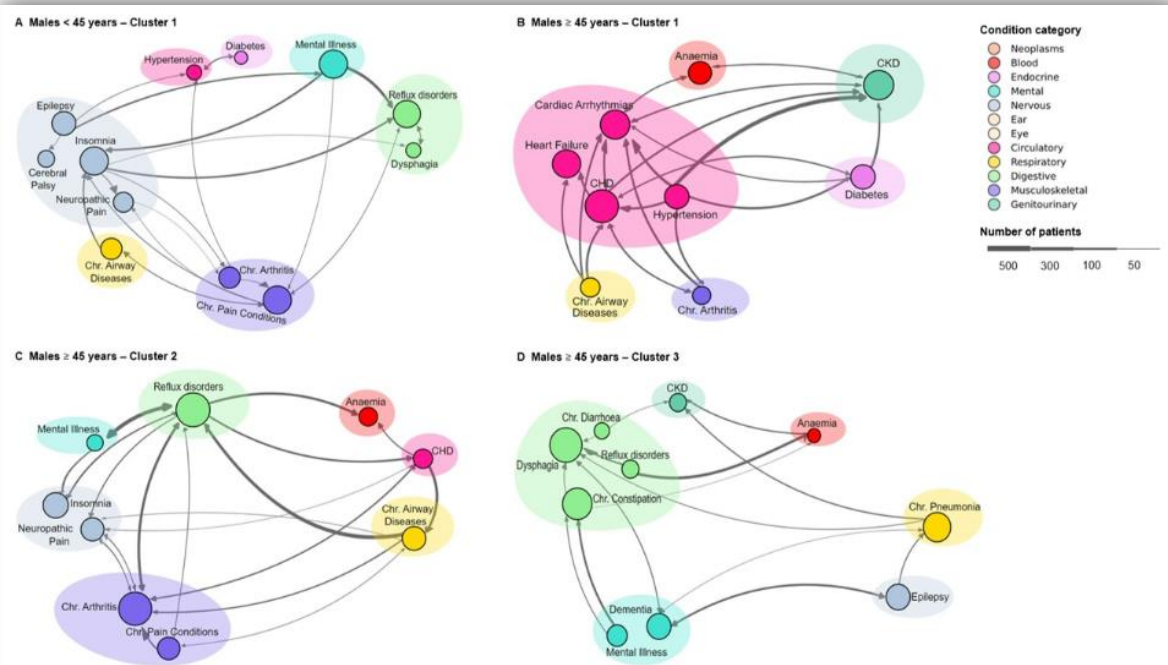
<https://www.decode-project.org/>



Funding: DECODE(NIHR203981) is funded by the NIHR AI for Multiple Long-term Conditions (AIM) Programme.

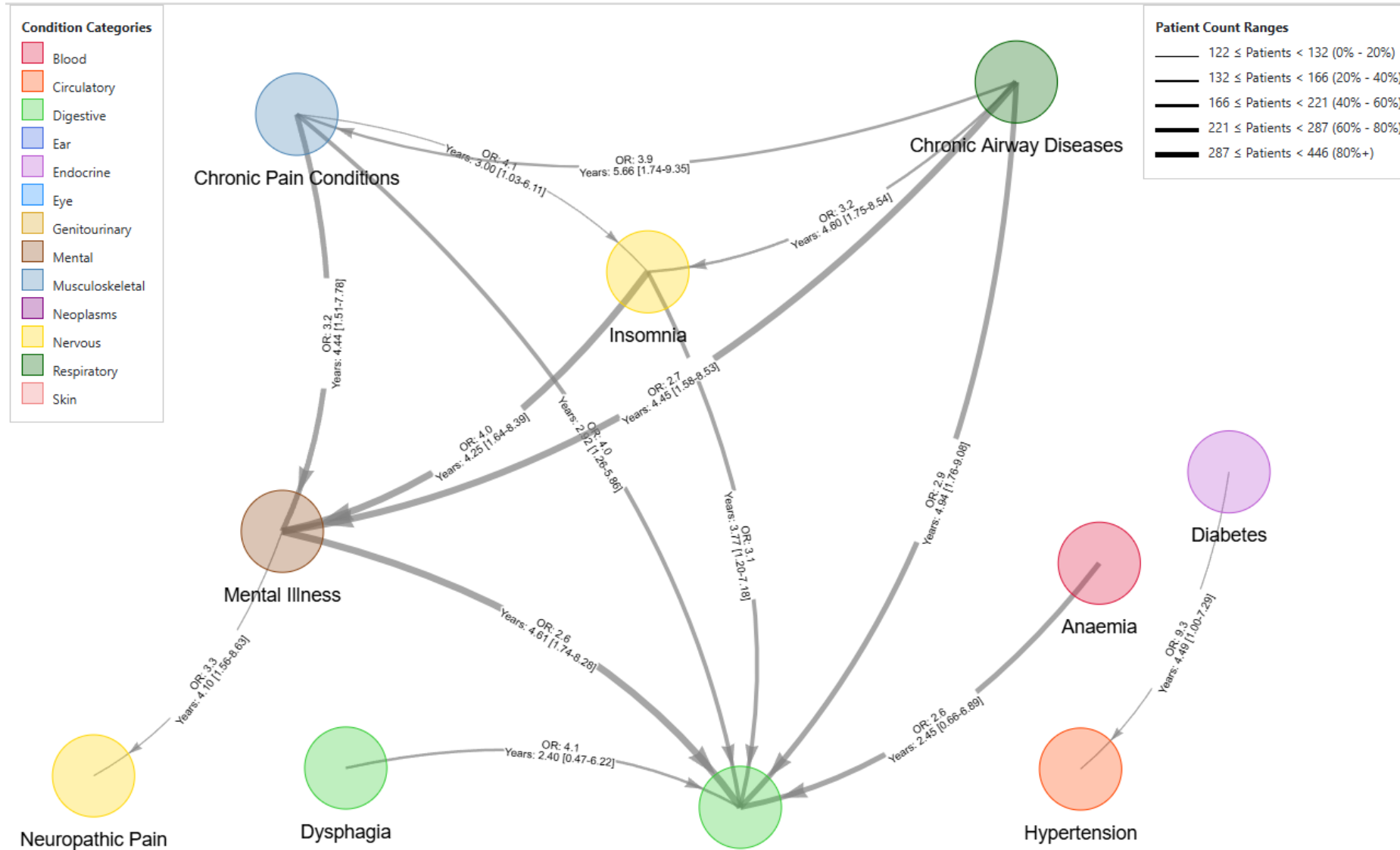
FUNDED BY
NIHR | National Institute for
Health and Care Research





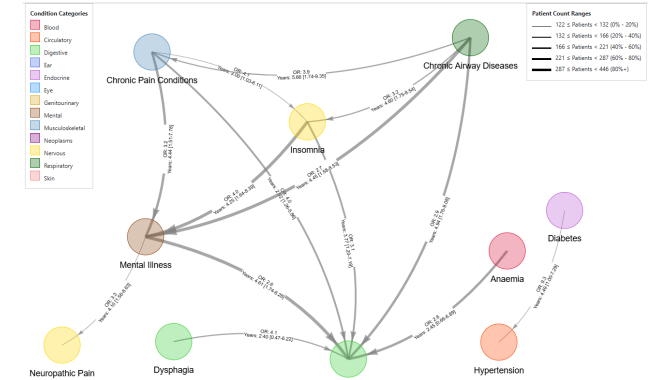
What was AI used for?

- Clusters of conditions
- Temporal modelling of disease trajectories
- Predicting length of stay using machine learning
- Fairness analysis - bias management



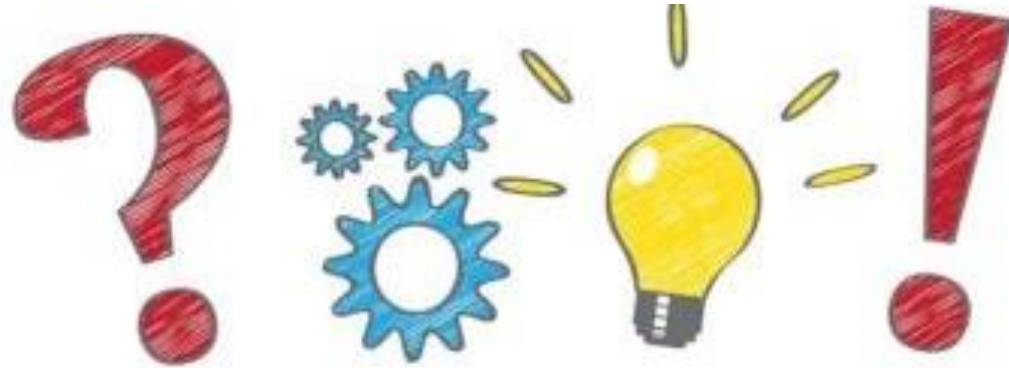
(a) Sex: Female. Age group: <45. Total patients with diagnoses in this group: 5346.

- Most prevalent conditions
- Statistically significant pairs
- Temporal Trajectories



Females under 45 years		
Combination	No. of conditions	Minimum Combination Prevalence %
Chronic Airway Diseases + Mental Illness + Reflux Disorders	3	4.43
Insomnia + Mental Illness + Reflux Disorders	3	3.50
Chronic Pain Conditions + Mental Illness + Reflux Disorders	3	3.40
Insomnia + Chronic Airway Diseases + Reflux Disorders	3	3.03
Insomnia + Chronic Airway Diseases + Mental Illness	3	3.03
Insomnia + Chronic Airway Diseases + Mental Illness + Reflux Disorders	4	3.03
Chronic Pain Conditions + Chronic Airway Diseases + Reflux Disorders	3	2.88

Thank you!



Please get in touch if you are interested in exploring collaboration or trying out the tools.

Georgina Cosma
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